TMUA PRACTICE TEST SERIES

TMUA Topic Test 4

Logic and Proofs

- 20 multiple-choice questions
- Recommended time: 75 minutes
- Paper 2 styled
- Challenging reasoning and proof-audit set

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

TMUA Topic Test 4 — Logic and Proofs

Complete worked solutions for the 20-question TMUA Paper 2 styled Logic and Proofs topic test. Question wording, answer choices, diagrams and solutions have been checked against one canonical question set.

Answer key

1 **C**	2 **F**	3 **C**	4 **B**	5 **B**	6 **C**	7 **D**	8 **E**	9 **C**	10 **D**
11 **E**	12 **D**	13 **C**	14 **H**	15 **C**	16 **D**	17 **F**	18 **A**	19 **D**	20 **B**

TMUA PAPER 2

Conditions and Implications

Necessary and sufficient conditions, biconditionals, divisibility and rigorous interpretation.

Q1

Necessary and Sufficient Conditions

Answer C

Question 1

The polynomial

$$P(x) = x^3 + (a-3)x^2 + (b-3a)x - 3b$$

has exactly two distinct real roots. Which condition is necessary and sufficient?

A $a^2 - 4b = 0$

B $9 + 3a + b = 0$

C $(a^2 - 4b = 0 \text{ and } 9 + 3a + b \neq 0)$ or $(9 + 3a + b = 0 \text{ and } a^2 - 4b > 0)$

D $a^2 - 4b \geq 0 \text{ and } 9 + 3a + b = 0$

E None of the above

WORKED SOLUTION

Factor first:

$$P(x) = (x-3)(x^2 + ax + b).$$

There are exactly two distinct real roots in precisely two mutually exclusive cases.

1. The quadratic has one repeated real root, but that root is not 3. This requires

$$a^2 - 4b = 0 \quad \text{and} \quad 9 + 3a + b \neq 0.$$

2. The quadratic has two distinct real roots and one of them is 3. This requires

$$9 + 3a + b = 0 \quad \text{and} \quad a^2 - 4b > 0.$$

If both $9 + 3a + b = 0$ and $a^2 - 4b = 0$, the polynomial is $(x-3)^3$, so it has only one distinct root.

Hence the exact necessary-and-sufficient condition is option $\boxed{\text{C}}$.

Q2

Logical Conditions and Domains

Answer **F**

Question 2

Which of the following statements are true?

I. For real x, $x^2 > 4$ is sufficient for $x > 2$.
II. For an integer n, being even is necessary and sufficient for n^2 to be even.
III. Being a rectangle is necessary, but not sufficient, for being a square.
IV. For real x, $x > 3$ is necessary for $x^2 > 9$.
V. For an integer n, $n < 19.5$ is necessary and sufficient for $n \leq 19$.

- **A** I, II and III
- **B** II and III only
- **C** II and V only
- **D** III and V only
- **E** I and IV only
- **F** II, III and V only

WORKED SOLUTION

I is false: $x = -3$ satisfies $x^2 > 4$, but not $x > 2$.

II is true: an integer and its square have the same parity.

III is true: every square is a rectangle, but not every rectangle is a square.

IV is false: $x = -4$ gives $x^2 > 9$ without $x > 3$.

V is true because, for integers, $n < 19.5$ and $n \leq 19$ describe exactly the same set.

Therefore II, III and V only are true: $\boxed{\text{F}}$.

Q3

Implication and Contrapositive

Answer C

Question 3

Let I, II, III, IV be statements. Suppose

$$I \Rightarrow II, \qquad II \Rightarrow III, \qquad IV \Rightarrow \neg III, \qquad \neg I \Rightarrow II.$$

Suppose II is true. What can be concluded about the other statements?

- **A** I true; III true; IV true
- **B** I could be either; III true; IV could be either
- **C** I could be either; III true; IV false
- **D** I false; III true; IV false
- **E** I true; III false; IV false

WORKED SOLUTION

Since $II \Rightarrow III$ and II is true, III is true.

The contrapositive of $IV \Rightarrow \neg III$ is $III \Rightarrow \neg IV$. Therefore IV is false.

Nothing forces I: if I is true, $I \Rightarrow II$ is satisfied; if I is false, $\neg I \Rightarrow II$ is satisfied.

Thus I could be either, III is true and IV is false. The answer is $\boxed{\text{C}}$.

Q4

Biconditional Proof

Answer **B**

Question 4

Let I and II be statements. You are asked to prove I if and only if II. Which pair of implications does **not** prove the biconditional?

- **A** $I \Rightarrow II$ and $II \Rightarrow I$
- **B** $II \Rightarrow \neg I$ and $I \Rightarrow \neg II$
- **C** $\neg I \Rightarrow \neg II$ and $\neg II \Rightarrow \neg I$
- **D** $\neg II \Rightarrow \neg I$ and $II \Rightarrow I$

WORKED SOLUTION

A is the direct two-way proof.

C consists of the contrapositives of $II \Rightarrow I$ and $I \Rightarrow II$, so it proves both directions.

D gives $I \Rightarrow II$ through its contrapositive $\neg II \Rightarrow \neg I$, together with $II \Rightarrow I$.

B instead says that I and II cannot be true together; it does not prove equivalence. Hence $\boxed{\text{B}}$.

Q5

Conditions for Real Solutions

Answer B

Question 5

Which condition on a is necessary and sufficient for the equations

$$y = x - 4, \qquad x^2 - 2y^2 = a$$

to have at least one real solution?

- A $a < 32$
- B $a \leq 32$
- C $a > 32$
- D $a \geq 32$
- E $a < 16$
- F $a \leq 16$
- G $a > 16$
- H $a \geq 16$

WORKED SOLUTION

Substitute $y = x - 4$:

$$a = x^2 - 2(x-4)^2 = -x^2 + 16x - 32 = 32 - (x-8)^2.$$

The expression $32 - (x-8)^2$ takes every real value less than or equal to 32, and no value greater than 32.

Therefore the equations have a real solution exactly when $a \leq 32$. The answer is $\boxed{\mathrm{B}}$.

Q6

Counterexamples and Divisibility

Answer C

Question 6

Consider the statement:

If n is an integer and n^2 is divisible by 432, then n is divisible by 72.

How many counterexamples are there in the range $100 \leq n \leq 1000$?

- **A** 11
- **B** 12
- **C** 13
- **D** 14
- **E** 25

WORKED SOLUTION

Since

$$432 = 2^4 \cdot 3^3,$$

the condition $432 \mid n^2$ requires

$$v_2(n) \geq 2, \qquad v_3(n) \geq 2.$$

Thus $36 \mid n$. The statement fails exactly for multiples of 36 that are not multiples of 72.

In the given range, the number of multiples of 36 is

$$\left\lfloor \frac{1000}{36} \right\rfloor - \left\lfloor \frac{99}{36} \right\rfloor = 27 - 2 = 25.$$

The number of multiples of 72 is

$$\left\lfloor \frac{1000}{72} \right\rfloor - \left\lfloor \frac{99}{72} \right\rfloor = 13 - 1 = 12.$$

Hence there are $25 - 12 = 13$ counterexamples. The answer is $\boxed{\text{C}}$.

Q7

Necessary and Sufficient Conditions

Answer **D**

Question 7

Assume that both infinite expressions below converge to real values. Find the necessary and sufficient condition on a for them to be equal:

$$\sqrt{a-\sqrt{a-\sqrt{a-\cdots}}} = \cfrac{1}{a-\cfrac{1}{a-\cfrac{1}{a-\cdots}}}.$$

- **A** True for every real a
- **B** $a=0$
- **C** $a=1$
- **D** $a=2$
- **E** $a=\sqrt{2}$

WORKED SOLUTION

Let the common value be x. Periodicity of the two convergent expressions gives

$$x=\sqrt{a-x}, \qquad x=\frac{1}{a-x}.$$

The first equation gives $a-x=x^2$. Substituting into the second gives

$$x=\frac{1}{x^2} \implies x^3=1.$$

Since the radical is non-negative, $x=1$. Then $1=a-1$, so $a=2$.

Conversely, when $a=2$, both standard convergent iterations tend to 1. Hence $a=2$ is necessary and sufficient, and the answer is $\boxed{\mathrm{D}}$.

TMUA PAPER 2

Counterexamples and Proof Strategy

Circle-theorem proof routes, logical equivalence, integrals and quantified counterexamples.

Q8

Proof Strategy in Circle Geometry

Answer E

Question 8

Points A, B, X, Y lie on one circle, with X and Y in the same segment of chord AB. The goal is to prove

$$\angle AXB = \angle AYB.$$

Assume each theorem below has already been established independently. Which listed theorems, used separately, are each sufficient to prove the result?

I. The angle at the centre is twice the angle at the circumference standing on the same arc.
II. The angle between a tangent and chord AB equals the angle in the alternate segment.
III. Opposite angles of a cyclic quadrilateral sum to 180°.

A None

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H All three

WORKED SOLUTION

I is sufficient: both $\angle AXB$ and $\angle AYB$ are half the same central angle subtended by chord AB .

II is also sufficient: after drawing a tangent at A, both angles equal the same angle between that tangent and chord AB.

III alone only gives a supplementary relationship between opposite angles of a cyclic quadrilateral; it does not by itself establish the required equality.

Therefore I and II only are sufficient. The answer is $\boxed{\text{E}}$.

Q9

Errors in Equivalence

Answer C

Question 9

A student attempts to solve $2\log_5 x = 3$. The symbols $\Rightarrow$ and $\iff$ are intended literally.

$$\begin{aligned} 2\log_5 x = 3 &\iff \log_5(x^2) = 3 && \text{(Line 2)} \\ &\iff x^2 = 125 && \text{(Line 3)} \\ &\Rightarrow x = \sqrt{125} = 5\sqrt{5} && \text{(Line 4)} \end{aligned}$$

On which lines does the working contain an algebraic or logical error?

- **A** Lines 2 and 3
- **B** Lines 3 and 4
- **C** Lines 2 and 4
- **D** Lines 2, 3 and 4

WORKED SOLUTION

Line 2 is not an equivalence. The original expression requires $x > 0$, whereas $\log_5(x^2)$ is also defined for negative x.

Line 3 is a valid equivalence on the domain of $\log_5(x^2)$.

Line 4 is false as an implication because $x^2 = 125$ permits both $x = \sqrt{125}$ and $x = -\sqrt{125}$.

Thus the errors are on lines 2 and 4, so the answer is $\boxed{\text{C}}$.

Q10

Necessary Conditions for Integrals

Answer D

Question 10

Let f be integrable on $[-1,1]$ and satisfy $f(x) \le 0$ for every $x \ge 0$. Which condition is necessary for

$$\int_{-1}^{1} f(x)\,dx > 0?$$

- **A** $f(x) = -f(-x)$ for all x
- **B** $f(0) = 0$
- **C** $f(x) \ge 0$ for all $x < 0$
- **D** $\int_{-a}^{0} f(x)\,dx > 0$ for some $a \in (0,1]$
- **E** $f(x) = f(-x)$ for all x
- **F** $f(-1) > 0$

WORKED SOLUTION

Because $f(x) \le 0$ on $[0,1]$,

$$\int_{0}^{1} f(x)\,dx \le 0.$$

But the total integral is positive, so

$$\int_{-1}^{0} f(x)\,dx = \int_{-1}^{1} f(x)\,dx - \int_{0}^{1} f(x)\,dx > 0.$$

Thus D is automatically satisfied by taking $a = 1$. None of the pointwise symmetry or sign conditions in the other options is forced.

The answer is $\boxed{\text{D}}$.

Q11

Counterexamples to a Biconditional

Answer E

Question 11

For a positive integer p, consider the claim:

$$p \text{ is prime} \iff p = 4k+3 \text{ for some non-negative integer } k.$$

Which of the following values are counterexamples to the biconditional?

I. $p = 13$
II. $p = 27$
III. $p = 19$
IV. $p = 21$

A None

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and IV only

WORKED SOLUTION

A biconditional fails whenever its two sides have different truth values.

- 13 is prime but is not of the form $4k+3$: counterexample.
- $27 = 4 \cdot 6 + 3$ but is not prime: counterexample.
- $19 = 4 \cdot 4 + 3$ and is prime: not a counterexample.
- 21 is neither prime nor of the form $4k+3$: both sides are false, so it is not a counterexample to the biconditional.

Therefore I and II only are counterexamples. The answer is $\boxed{\text{E}}$.

Q12

Negating a Universal Implication

Answer D

Question 12

Consider the statement:

For every prime p, if $p = 4k+1$ for some positive integer k, then there are positive integers a, b such that $p = a^2 + b^2$.

Which statement, if true, would prove the claim false?

A For every prime p, if $p = a^2 + b^2$, then $p = 4k+1$.

B For every prime p, if $p \neq a^2 + b^2$ for all positive a, b, then $p = 4k+1$.

C For every prime p, $p = 4k+1$ if and only if $p = a^2 + b^2$ for some positive a, b.

D There exists a prime p such that $p = 4k+1$ and $p \neq a^2 + b^2$ for all positive a, b.

E There exists a prime p such that $p = a^2 + b^2$ for some positive a, b, or $p = 4k+1$.

WORKED SOLUTION

A universal implication

$$\forall p\ (A(p) \Rightarrow B(p))$$

is false exactly when there exists at least one p for which $A(p)$ is true and $B(p)$ is false.

Option D states precisely that there is a prime of the form $4k+1$ that is not expressible as a sum of two positive squares. Hence $\boxed{\text{D}}$.

TMUA PAPER 2

Proof Auditing and Advanced Reasoning

Geometry proofs, index laws, contradiction, graph logic, induction and extraneous roots.

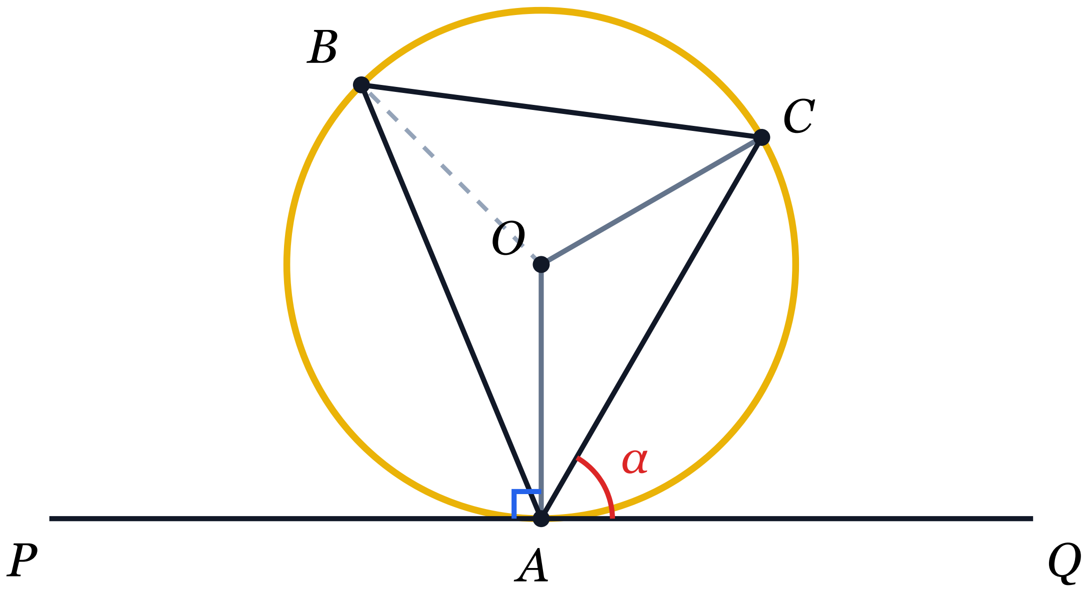
Q13

First Error in a Geometric Proof

Answer C

Question 13

Consider the following attempted proof of the alternate segment theorem.



I. A tangent touches a circle at A, and B, C lie on the circle. The angle between CA and the tangent is $\alpha°$.
II. Radii OA and OC are drawn. The angle between OA and the tangent is $90°$.
III. Therefore $\angle BAO = (90 - \alpha)°$.
IV. Since triangle OBA is isosceles, $\angle OBA = (90 - \alpha)°$.
V. Hence $\angle AOB = 2\alpha°$.
VI. Therefore $\angle ACB = \alpha°$.

Where is the first error?

- **A** The proof is correct
- **B** Line II
- **C** Line III
- **D** Line IV
- **E** Line V
- **F** Line VI

WORKED SOLUTION

Line II is correct: a radius is perpendicular to the tangent at the point of contact.

However, the angle related to α is $\angle CAO$, not $\angle BAO$. The position of B has not yet been related to the tangent or to CA.

Thus line III is the first unsupported step. The answer is $\boxed{\text{C}}$.

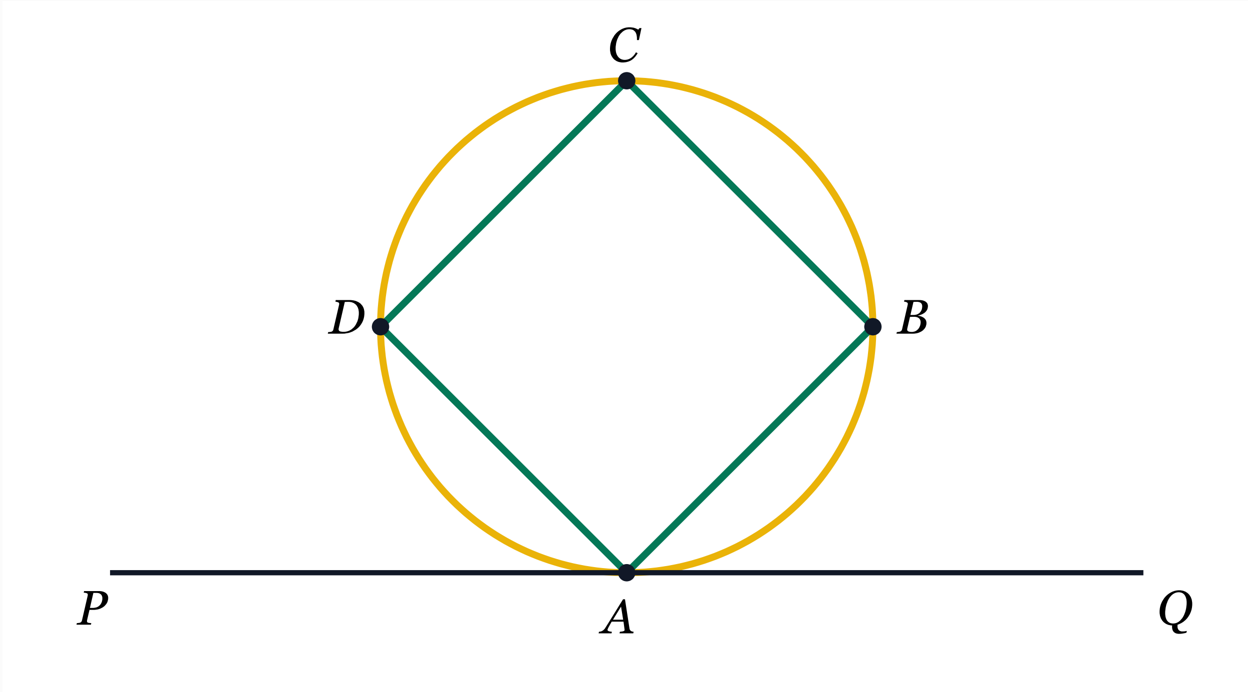
Q14

Necessary Conditions in Geometry

Answer **H**

Question 14

The diagram shows a cyclic quadrilateral $ABCD$ and the tangent PQ at A.



Which conditions are necessary if $ABCD$ is a rhombus?

I. $\angle PAD = 45^\circ$
II. $CA = 2r$, where r is the radius of the circle
III. $CD = CB$

- **A** None
- **B** I only
- **C** II only
- **D** III only
- **E** I and II only
- **F** I and III only
- **G** II and III only
- **H** All three

WORKED SOLUTION

A cyclic rhombus must be a square: opposite angles of a rhombus are equal, while opposite angles of a cyclic quadrilateral are supplementary. Therefore every angle is 90°.

III is immediate because every side of a rhombus is equal.

Since $\angle ABC = 90^\circ$, chord AC is a diameter, so $CA = 2r$. Thus II is necessary.

In the square, diagonal AC bisects $\angle DAB$, giving $\angle DAC = 45^\circ$. The tangent at A is perpendicular to diameter AC, so $\angle PAC = 90^\circ$. Hence $\angle PAD = 45^\circ$, making I necessary as well.

All three conditions are necessary. The answer is $\boxed{\text{H}}$.

Q15

First Error in Index Laws

Answer C

Question 15

A student tries to solve

$$\frac{2^{18x}}{2^{3x^2} \cdot 4^6} > 1.$$

Their working is:

I. $2^{18x-3x^2} \cdot 4^{-6} > 1$
II. $8^{6x-3x^2} \cdot 4^{-6} > 1$
III. $8^{6x-x^2-3} > 1$
IV. $6x - x^2 - 3 > 0$
V. The critical values are $3 \pm \sqrt{6}$.
VI. Therefore $3 - \sqrt{6} < x < 3 + \sqrt{6}$.

Where is the first error?

A The working is correct

B Line I

C Line II

D Line III

E Line IV

F Line V

G Line VI

WORKED SOLUTION

Line I is correct.

To rewrite a power of 2 as a power of $8 = 2^3$, the exponent must be divided by 3:

$$2^{18x-3x^2} = 8^{6x-x^2},$$

not 8^{6x-3x^2}. Therefore line II is the first error, and the answer is $\boxed{\text{C}}$.

Q16

First Error in Proof by Contradiction

Answer **D**

Question 16

Consider the following attempted proof by contradiction that $\sqrt{12}$ is irrational.

I. Suppose $\sqrt{12} = a/b$, where integers a, b have no common divisor.
II. $12 = a^2/b^2$.
III. $12b^2 = a^2$.
IV. Since $12 \mid a^2$, it follows that $12 \mid a$, so $a = 12c$.
V. Then $144c^2 = 12b^2$, so $b^2 = 12c^2$.
VI. Using the same logic, $b = 12d$.
VII. Then a and b have a common divisor, a contradiction.

Where is the first error?

- **A** The proof is correct
- **B** Line I
- **C** Line II
- **D** Line IV
- **E** Line V
- **F** Line VI
- **G** Line VII

WORKED SOLUTION

Lines I–III are valid.

Line IV is false. From $12 \mid a^2$, one may conclude that $6 \mid a$, but not necessarily that $12 \mid a$. For instance, $a = 6$ gives $12 \mid a^2 = 36$, yet $12 \nmid 6$.

Thus line IV is the first error, and the answer is $\boxed{\text{D}}$.

Q17

Sufficiency, Necessity and Graphs

Answer F

Question 17

Consider the graphs

$$y = mx - 10 \qquad \text{and} \qquad y = \log_2 x.$$

Relative to the statement "the two graphs have at least one intersection", consider:

I. $m < 0$
II. $m > 10$

Which classification is correct?

- **A** Both are necessary and sufficient
- **B** Both are necessary
- **C** Both are sufficient
- **D** One is sufficient and the other is necessary
- **E** Only I is necessary
- **F** Only I is sufficient
- **G** Neither is necessary nor sufficient

WORKED SOLUTION

Let

$$h(x) = mx - 10 - \log_2 x,$$

so intersections correspond to zeros of h.

If $m < 0$, then as $x \to 0^+$, $h(x) \to +\infty$, while as $x \to \infty$, $h(x) \to -\infty$. Continuity guarantees an intersection. Thus I is sufficient.

I is not necessary because some positive slopes also produce intersections.

II is not necessary because all negative slopes work. It is not sufficient either. For example, when $m = 1024$, the maximum of

$$\log_2 x - mx + 10$$

occurs at $x = 1/(m \ln 2)$ and is negative, so the logarithmic curve never reaches the line.

Only I is sufficient. The answer is $\boxed{\text{F}}$.

Q18

Checking an Induction Proof

Answer A

Question 18

Consider the following induction proof that $3 \cdot 7^n + 6$ is divisible by 9 for every non-negative integer n.

I. For $n = 0$, $3 \cdot 7^0 + 6 = 9$, which is divisible by 9.
II. Assume $3 \cdot 7^k + 6 = 9M$ for some non-negative integer k and integer M.
III. $3 \cdot 7^{k+1} + 6 = 7(3 \cdot 7^k + 6) - 36$.
IV. Therefore $3 \cdot 7^{k+1} + 6 = 7(9M) - 36 = 9(7M - 4)$.
V. Thus the result holds for $k + 1$, and hence for every non-negative integer n.

Where is the first error, if any?

- **A** The proof is correct
- **B** Line I
- **C** Line II
- **D** Line III
- **E** Line IV
- **F** Line V

WORKED SOLUTION

The base case $n = 0$ is correct.

The induction hypothesis is correctly stated for $n = k$.

Line III is also correct:

$$7(3 \cdot 7^k + 6) - 36 = 21 \cdot 7^k + 6 = 3 \cdot 7^{k+1} + 6.$$

Substitution of the induction hypothesis gives a multiple of 9, completing the step. Therefore the proof is correct and the answer is $\boxed{\text{A}}$.

Q19

Counterexamples and Exact Quantifiers

Answer D

Question 19

Which function is a counterexample to the statement:

If a function has exactly three distinct real roots, then it is a cubic polynomial.

- **A** $\sin x$ on $0 < x < 2\pi$
- **B** $(x-1)(x-2)(x-3)(x-4)$
- **C** $\cos x$ on $0 < x < 2\pi$
- **D** $(x-1)^2(x-2)(x-3)$
- **E** $(x+1)(x^2-x+5)$
- **F** $(x-3)(x+1)(x-4)$

WORKED SOLUTION

A counterexample must have exactly three distinct real roots but must not be cubic.

Option D is quartic, yet its distinct real roots are exactly $1, 2, 3$. Therefore it satisfies the hypothesis and violates the conclusion.

B has four distinct roots, while F is cubic. Hence the unique answer is $\boxed{\text{D}}$.

Q20

Errors Caused by Squaring

Answer **B**

Question 20

A student attempts to solve

$$\sqrt{x+3} = 3x - 1.$$

Their working uses equivalence signs:

I. $\sqrt{x+3} = 3x - 1 \iff x + 3 = (3x-1)^2$
II. $x + 3 = 9x^2 - 6x + 1$
III. $0 = 9x^2 - 7x - 2$
IV. $0 = (x-1)(9x+2)$
V. So $x = 1$ and $x = -\frac{2}{9}$.

Which assessment is correct?

- **A** Both answers are correct
- **B** Only $x = 1$ is correct, and the first error is line I
- **C** Only $x = 1$ is correct, and the first error is line II
- **D** Only $x = 1$ is correct, and the first error is line IV
- **E** Neither answer is correct, and the first error is line I
- **F** Neither answer is correct, and the first error is line IV

WORKED SOLUTION

Squaring is valid as a one-way implication, but not automatically as an equivalence. The squared equation can admit values for which $3x - 1 < 0$, while the principal square root is non-negative.

Both algebraic candidates solve the squared equation. Substitution into the original gives:

$$x = 1: \quad \sqrt{4} = 2 = 3(1) - 1,$$

so $x = 1$ works. But for $x = -2/9$, the right-hand side is negative, so it cannot equal a square root.

Thus only $x = 1$ is correct, and the first error is the false equivalence in line I. The answer is $\boxed{\text{B}}$.

solution PDF are retained in the relevant cards; embedded fallbacks are provided where possible.